UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

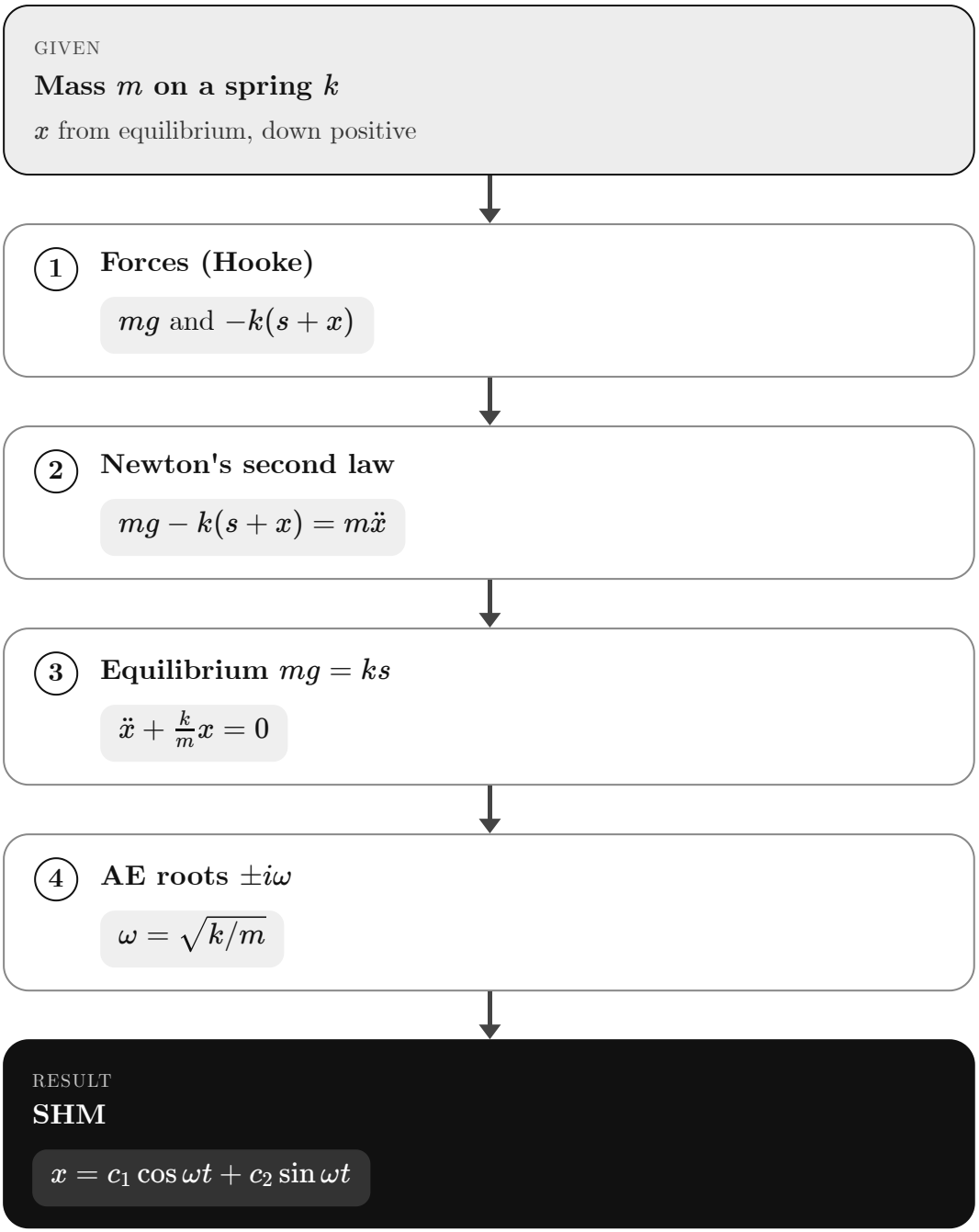
Class 015 · Applications of LDEs: simple harmonic oscillation · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S015·11	Unit 2 deck 015 (“Applications of DE – SHM”), slide 11 (rendered .pptx)
T015 00:11:48	Unit 2 AV-summary transcript “Application problems on linear differential equations – Simple harmonic oscillation” (16.3 min), segment start
CW U2 P37	Unit 2 class-work problem 37

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7. Every spring problem of this class, side by side
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1 The spring–mass model and simple harmonic motion



Remark. Steps 4–5 are the complex-root case of class 002.

Figure F1.1. From forces to simple harmonic motion ESA ★

S015·6 · S015·7 · S015·8 · T015 00:04:30 · T015 00:05:33

The quantities of simple harmonic motion

Quantity	Formula	12 lb example	8 lb example
Angular frequency	$\omega = \sqrt{k/m}$	8 rad/s	$4\sqrt{3}$ rad/s
Amplitude	$A = \sqrt{c_1^2 + c_2^2}$	$\frac{5}{12}$ ft	$\frac{5}{12}$ ft
Period	$T = \frac{2\pi}{\omega}$	$\frac{\pi}{4}$ s	$\frac{\pi}{2\sqrt{3}}$ s
Frequency	$f = \frac{1}{T}$	$\frac{4}{\pi}$ Hz	$\frac{2\sqrt{3}}{\pi}$ Hz
Max velocity	$A\omega$	$\frac{10}{3}$ ft/s	$\frac{5}{\sqrt{3}}$ ft/s

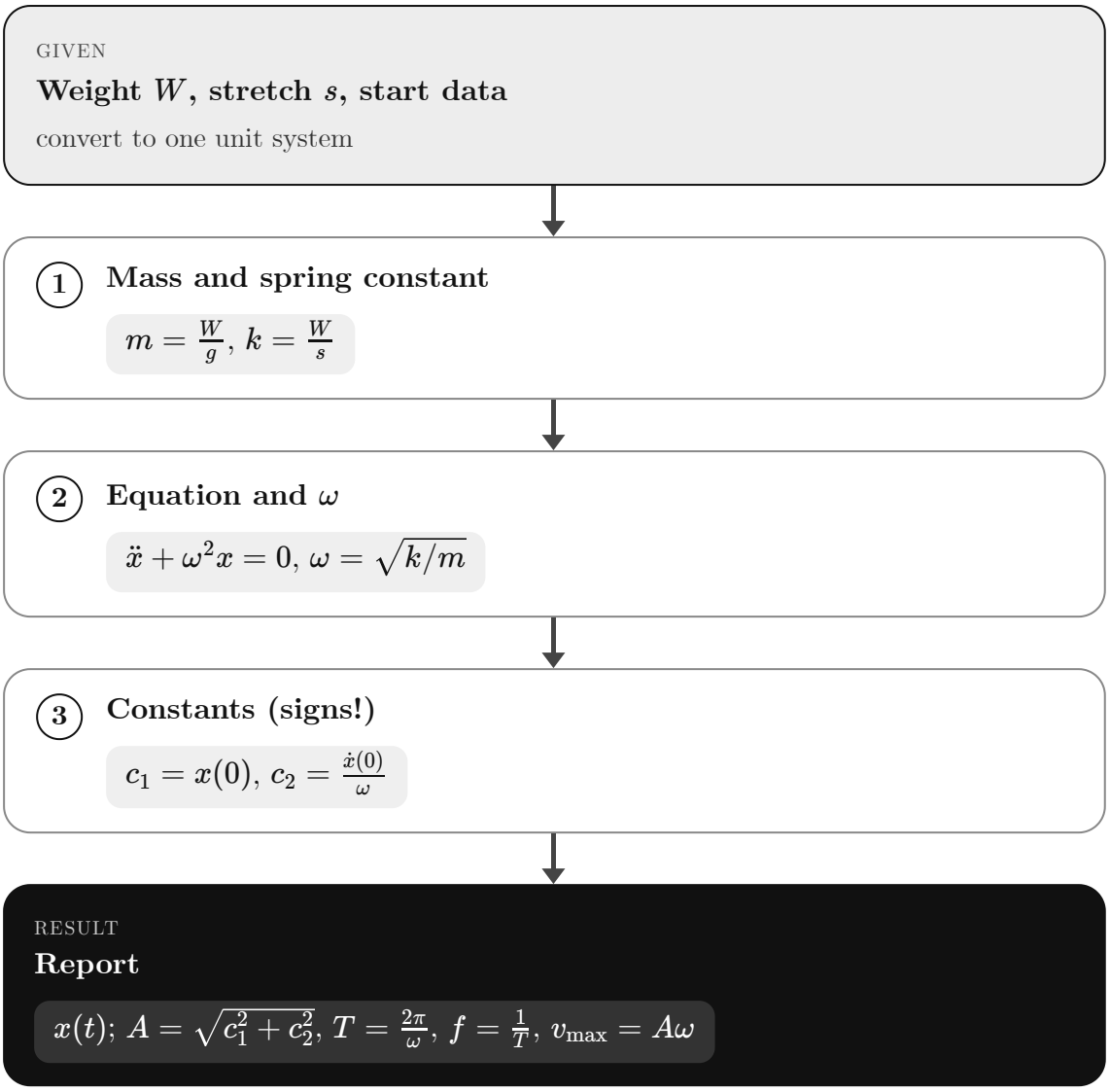
S015·7 · S015·8 · S015·9 · S015·22

Spring–mass system and LCR circuit

Spring–mass	LCR circuit (class 014)
$m\ddot{x} + c\dot{x} + kx = F(t)$	$L\ddot{Q} + R\dot{Q} + \frac{Q}{C} = E(t)$
mass m	inductance L
damping coefficient c	resistance R
spring constant k	$\frac{1}{C}$
displacement x, velocity $\dot{x}$	charge Q, current I
$\omega = \sqrt{k/m}$	$\omega = \frac{1}{\sqrt{LC}}$

added for clarity S015·4 · T015 00:00:32

2 Solving spring problems

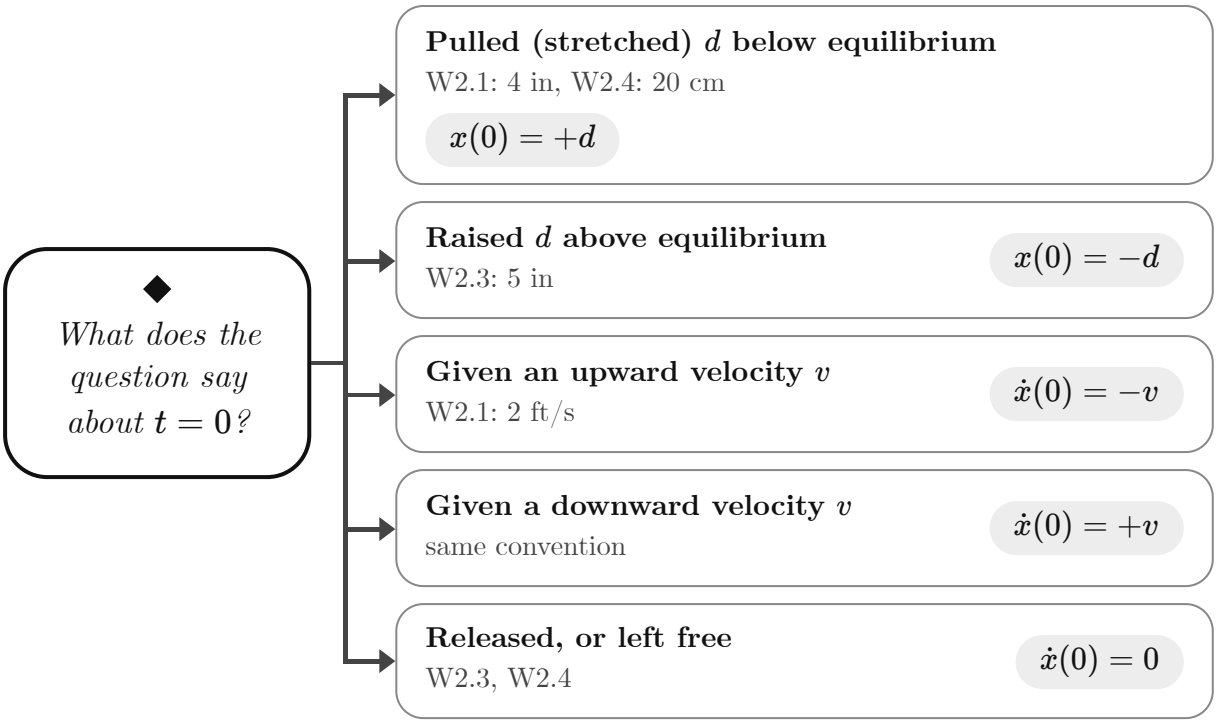


Remark. W2.1, W2.3 and W2.4 all follow these four steps.

Exam trap. Inches to feet before using $g = 32$ ft/s².

Figure F2.1. Solving a spring problem ESA ★★

S015·10 · S015·11 · S015·12 · S015·13 · S015·14 · T015 00:09:25 · T015 00:12:41



Remark. The convention is stated in the lecture and on slide 015·20.

Figure F2.2. Turning the words into initial conditions (down positive) ESA ★★

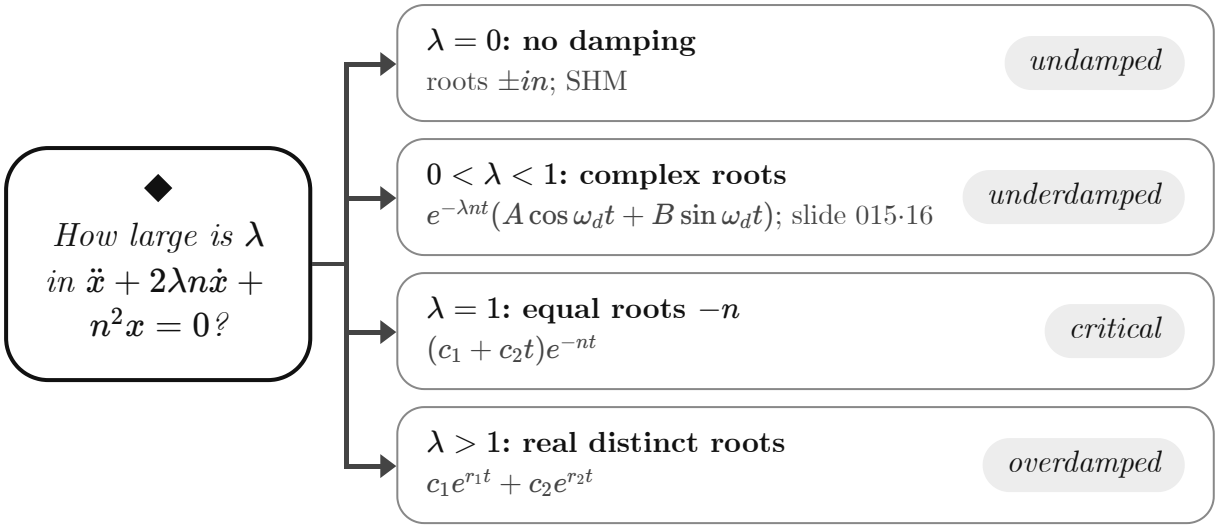
S015·13 · S015·14 · S015·20 · T015 00:14:07 · T015 00:14:24

Every spring problem of this class, side by side

Problem	m, k	Equation	$x(0)$, $\dot{x}(0)$	$x(t)$
12 lb, 6 in (W2.1)	$\frac{3}{8}$, 24	$\ddot{x} + 64x = 0$	$\frac{1}{3}$, -2	$\frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$
8 lb, $k = 12$ (W2.3)	$\frac{1}{4}$, 12	$\ddot{x} + 48x = 0$	$-\frac{5}{12}$, 0	$-\frac{5}{12}\cos 4\sqrt{3}t$
CW P37 (W2.4)	10, 1960	$\ddot{x} + 196x = 0$	0.2, 0	$0.2\cos 14t$

S015·10 · S015·17 · CW U2 P37

3 The damped oscillator



Remark. Only the first two oscillate.

Figure F3.1. Damped oscillator: which motion? ESA ★

ADDED FOR CLARITY S015·15 · S015·16

Undamped and damped oscillators

	Undamped	**Damped, $0 < \lambda < 1$**
Equation	$\ddot{x} + n^2x = 0$	$\ddot{x} + 2\lambda n\dot{x} + n^2x = 0$
Roots	$\pm in$	$-\lambda n \pm in\sqrt{1-\lambda^2}$
Frequency	n	$n\sqrt{1-\lambda^2}$ (smaller)
Amplitude	constant A	decays like $e^{-\lambda nt}$
$x(0) = 0,\ \dot{x}(0) = x_0$	$\frac{x_0}{n}\sin nt$	$\frac{x_0}{n\sqrt{1-\lambda^2}}e^{-\lambda nt}\sin(n\sqrt{1-\lambda^2}t)$

S015·7 · S015·15 · S015·16